

Deadlock and OS Scheduling Algorithms

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Deadlock

Deadlock is a condition in which two or more processes wait forever for resources held by each other.

- Occurs in Operating Systems
- Processes cannot continue execution
- Resources become blocked
- System performance decreases

Deadlock Diagram

P1 --> R1 --> waits for R2
P2 --> R2 --> waits for R1

Necessary Conditions for Deadlock

- Mutual Exclusion – Only one process can use a resource at a time
- Hold and Wait – Process holds one resource and waits for another
- No Preemption – Resource cannot be taken forcefully
- Circular Wait – Processes wait in a circular chain

Deadlock Prevention

Deadlock prevention removes one of the necessary conditions.

- Remove Mutual Exclusion
- Remove Hold and Wait
- Remove No Preemption
- Remove Circular Wait

Deadlock Avoidance

Deadlock avoidance prevents the system from entering an unsafe state.

- Uses Banker's Algorithm
- Checks safe state before resource allocation
- Avoids system crash

Deadlock Detection

The system detects deadlock after it occurs.

- Detect blocked processes
- Kill process
- Restart system
- Use resource preemption

CPU Scheduling

CPU Scheduling decides which process gets CPU execution time.

- Improves CPU utilization
- Reduces waiting time
- Supports multitasking
- Increases system efficiency

FCFS Scheduling

FCFS means First Come First Serve.

- First process arrives first executes
- Simple scheduling algorithm
- Easy to implement
- Long waiting time possible

P1 --> P2 --> P3

SJF Scheduling

SJF means Shortest Job First.

- Shortest burst time executes first
- Reduces average waiting time
- Efficient scheduling method
- Starvation may occur

P2 --> P3 --> P1

SRTF Scheduling

SRTF means Shortest Remaining Time First.

- Preemptive version of SJF
- Process with shortest remaining time executes
- Better response time
- More context switching

Priority Scheduling

Priority Scheduling executes the highest priority process first.

- Important processes execute first
- Priority assigned to each process
- Fast execution for critical tasks
- Starvation possible for low priority processes

P2 --> P1 --> P3

Round Robin Scheduling

Each process gets fixed CPU time called Time Quantum.

- Fair execution method
- Good for multitasking systems
- Each process gets equal CPU time
- Context switching overhead occurs

P1 --> P2 --> P3 --> P1

Multilevel Queue Scheduling

Processes are divided into multiple queues.

- System Queue
- Interactive Queue
- Batch Queue
- Different scheduling for different queues

Conclusion

Deadlock and Scheduling Algorithms are important concepts in Operating Systems.

- Help manage processes efficiently
- Improve CPU performance
- Support multitasking
- Ensure proper resource management